Heart Attack Prediction using Machine Learning

Mohamed Wed Eladham[a], Ali Bou NassifT[a], Mohammad AlShabi[b]
[a]Department of Computer Engineering, University of Sharjah, Sharjah, UAE; [b]Department of Mechanical and Nuclear Engineering, University of Sharjah, UAE

ABSTRACT

The most prevalent kind of cardiovascular illness is a heart attack, which may or may not have symptoms. The damage to the heart muscle increases with delayed treatment, which increases the risk of mortality. More than 10 million people die each year from heart attacks, and many of them may be avoided if heart attacks could be accurately predicted. To estimate the likelihood of suffering a heart attack, five different machine learning algorithms are used on the Public Health Heart Attack dataset. Several evaluation metrics, including accuracy, recall, precision, ROC curve, and F-score, were used to evaluate the models. All the models—MLP, RBF, SVM, KNN, and RF— achieved significant accuracies of more than 75%, with KNN having the greatest overall performance

Keywords: heart attack, machine learning, multilayer perceptron, support vector machine, k-nearest neighbors

1. INTRODUCTION

According to estimates from the World Health Organization (WHO), more than 17.9 million individuals pass away each year as a result of cardiovascular diseases. [1]. 80% of these fatalities are primarily the result of heart attacks. When one or more coronary arteries are clogged, the heart does not receive enough blood, which results in a heart attack. When cholesterol-rich fat deposits accumulate and form plaques, which are blood clots, the obstruction occurs. The main risk factors for developing these clots are an unhealthy lifestyle, using tobacco or alcohol, being overweight or obese and having high blood pressure or raised blood sugar levels. The American Heart Association lists the following as warning indicators: trouble breathing, chronic coughing , bruising in the feet and ankles, continual exhaustion, appetite loss, and mental confusion [2].

Heart attacks must be anticipated since they frequently occur silently. Artificial intelligence (AI) algorithms are now predicting illnesses including the Corona virus, diabetes, strokes, and heart attacks with a high degree of accuracy. This early disease prediction might stop the progression of the illness and make treatment much simpler. In this study, several machine learning models will be applied to identify patients who are more likely to experience a heart attack. For the predictions, a variety of classification techniques will be applied. The models will then be assessed using certain assessment indicators.

The literature used a variety of techniques to address the prediction of heart disease. With the same inputs as our work, Otoom et al. [3] used worn device measurements with Naïve Bayes classifier to forecast the likelihood of cardiac illnesses with 84.5% accuracy. Bharti et al. [4] predicted the likelihood of suffering a heart attack using the "Heart attack" dataset and several categorization approaches. To identify the outliers in the dataset, the isolation forest approach was applied. The algorithms utilized in training to obtain the predictions included kNN, SVM, Decision Trees, XGBoost, and Logistic Regression. KNN displayed the greatest result with 83.29% accuracy. Deep learning enabled the accuracy to rise to 94.2%. The dataset should have a gaussian distribution, it was determined.

Haq et al. [5] used a range of algorithms, such as Logistic Regression, KNN, and NN, as well as 3 feature selection methods to forecast the presence of cardiovascular diseases. The Logistic Regression method achieves results that were up to 89% correct, which was the best outcome.

Reddy et al. [6] effectively predicted the risk of heart disease using 10 machine learning classifiers using the "Cleveland heart disease" dataset. The most important characteristics were obtained using correlation-based FS, Chi squared, and Relief attribute assessors, and the models were trained using each approach independently. The sequential minimum optimization classifier on the chi-squared selected characteristics produced the maximum accuracy of 86.468%. It was determined that with the right attribute selection, good outcomes might be obtained.

* malshabi@sharjah.ac.ae

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This paper is structured as follows: In Section 2, the technological background is covered. The dataset is discussed and preprocessed in Section 3. In Section 4, the model design, evaluation measures selected to evaluate the model's performance, techniques employed, and pertinent model design parameters are described. The comments and conclusions concerning this study are offered in Section 5 as a last step.

2. RELATED WORK

2.1 Machine Learning

A subset of artificial intelligence known as machine learning (ML) imitates the actions of the human brain in order to help software programs anticipate outcomes more accurately without the need for explicit programming. It functions by building a mathematical model that is trained using practice data to learn and develop [7]. The outcomes of testing sets are then predicted using this model. Unsupervised learning and supervised learning are the two subcategories of machine learning. While unsupervised learning algorithms employ unlabeled data, supervised learning algorithms use data that has been labeled as input and output

2.2 Classification

Predicting the category of the inputs presented is the process of classification. It uses supervised learning to predict the function that connects the provided inputs to the discrete outputs [8]. Its primary goal is to determine which class does include the specific input.

2.3 Multi-layer Perceptron

A supervised learning approach called the multi-layer perceptron (MLP) can be used to train non-linear functions for classification or regression issues [8]. It works on a dataset with a goal output and a collection of characteristics X = x1, x2,... xm. It features more than one hidden layer with numerous computational neurons between the input and output layers, each of which has an activation function like ReLU or sigmoid. As a feedforward method, MLP adds the inputs together with their initial weights, calculates the sum, and propagates it to the following layer. A MLP example with several inputs, two hidden layers, and a single output is shown in Figure 1.

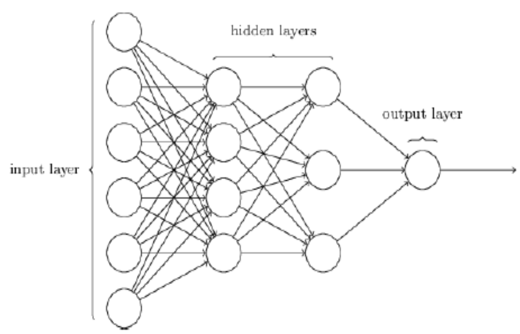


Figure 1. Multi-layer Perceptron example [9]

2.4 Radial-Basis Function Network

An input layer, a hidden layer, and an output layer make up the three layers of feed-forward neural networks that belong to the particular class of radial basis functions. This differs significantly from the majority of neural network topologies, which have several layers and generate nonlinearity by repeatedly using nonlinear activation functions [10]. The hidden layer is where computation takes place after receiving input data from the input layer. The Radial Basis Functions Neural Network's hidden layer is its most potent and distinctive feature compared to other neural networks. Prediction tasks like classification or regression are reserved for the output layer. Figure 2 shows the structure of the RBF model.

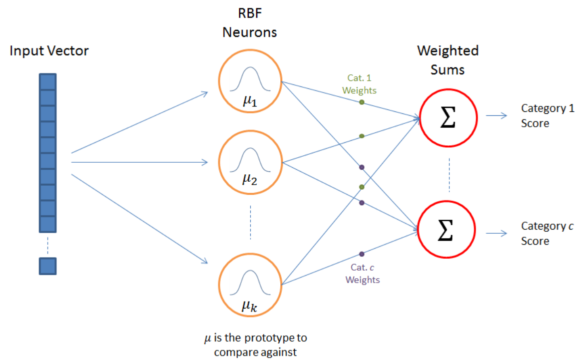


Figure 2. Radial Basis Function Network [11]

2.5 Support Vector Machine

The Support Vector Machine (SVM) method is one of the most used ones for classification. A linear classifier called SVM can distinguish between two groups of data [8]. It focuses primarily on computing the data margin, where it seeks to identify a hyperplane that divides training data classes by a margin. The greatest distance between this margin and each kind of data was considered while choosing it. As a result, the categorization error is reduced. Figure 3 shows an illustration of SVM in action.

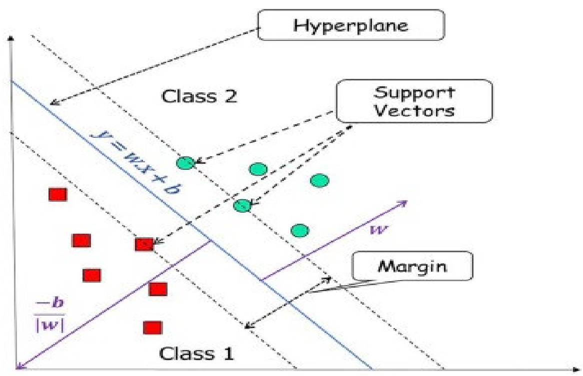


Figure 3. Support Victor Machine components [12]

2.6 k-Nearest Neighbors

K-Nearest Neighbors (kNN), one of the most straightforward supervised learning methods, is also helpful for classification and regression. [8]. Each example is stored, sorted into clusters based on similarities, and placed in the appropriate cluster after a verification of the testing data's properties. It measures the distance between the new location and the training spots. It then chooses the k points that are closest to the new point, averaging those points to decide which cluster the new point belongs in. Figure 4 is an illustration of a kNN example.

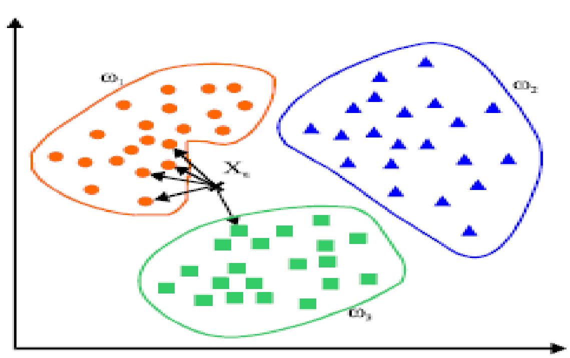


Figure 4. k-Nearest Neighbors example [13]

2.7 Random Forest

For both classification and regression, the Random Forest supervised learning method is used. In order to find answers for considerably more complicated issues, it employs ensemble learning, which combines classifiers [8]. The output of the random forest model is determined using the findings of several decision trees. The model's accuracy increases with the number of decision trees it contains. It uses the average of the outputs from the trees to estimate the outcome. Figure 5 shows an illustration of a random forest algorithm in action.

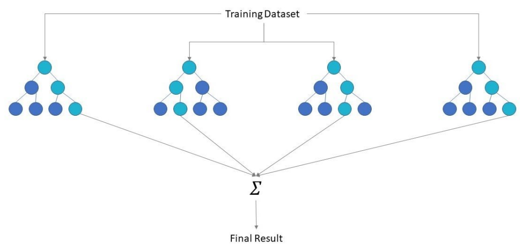


Figure 5. Random Forest example [14]

2.8 Evaluation Metrics

To evaluate the effectiveness and quality of the trained model, assessment metrics are used. Based on the following assessment measures, the best model in this study was chosen.

Confusion Matrix

A matrix of the form N*N, in which N represents the quantity of anticipated categories. Figure 6 illustrates the confusion matrix. The amount of the right and wrong predictions, broken down by class, make up the matrix's components. A TP, for instance, is the quantity of the correctly identified Positive class. In this instance, the quantity of cardiac illnesses. In a similar manner, the count of accurately anticipated absences in the Negative class—in this example, the absence of heart disease—represents a TN.

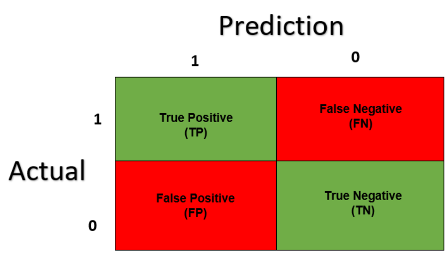


Figure 6. Confusion matrix [15]

Accuracy

Accuracy is a measure of the model's efficiency throughout all categories. It is useful when all categories are given equal weight. It is calculated by dividing the amount of accurate predictions by the total quantity of forecasts. The formula below can be used to get accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

The proportion of Positive samples that were properly categorized to all Positive samples. The precision demonstrates the model's accuracy in detecting Positive samples. Using the following equation, precision may be obtained:

$$Precision = \frac{TP}{TP + FP}$$

Recall

Recall is defined as the percentage of Positive samples that have been correctly labelled as Positive compared to all Positive samples. How effectively the model can recognize Positive samples is determined by recall. The recall increases as the number of positive samples is recognized. The following equation can be used to acquire recall:

$$Recall = \frac{TP}{TP + FN}$$

F-score

Given that it offers a harmonic mean for the recall and precision measurements in classification issues, the F-score would be the ideal option if the goal were to acquire the best accuracy and recall. F-score can be acquired using the following equation:

$$F - score = \frac{2TP}{2TP + FP + FN}$$

ROC Area

A plot that illustrates how effectively a classifying model works at every level of classification is called the receiver operating characteristic curve (ROC curve). On a ROC curve, TPR vs. FPR are represented for multiple classification parameters. Whenever the classification threshold is reduced, more objects are categorized as positive, which increases the quantity of both False Positives and True Positives. To produce the points on a ROC curve, we might evaluate a logistic regression model continuously using different classification criteria, yet this would be inefficient. However, we can retrieve this data using the effective sorting-based technique known as AUC.

"Area under the ROC Curve" is referred to by the acronym AUC. In other words, AUC determines the two-dimensional area beneath the whole ROC curve. AUC offers a comprehensive evaluation of performance throughout all available classification criteria. The likelihood that the system evaluates a randomly selected positive example higher than a randomly selected negative example is known as the AUC.

3. DATA DESCRIPTION AND PREPROCESSING

Because the quality of the data used to develop a machine learning model has a significant impact on how well it works, data preprocessing is essential. Data pretreatment entails purifying the eliminating outliers, damaged or empty data points, in addition to converting, resampling, and conducting feature selection.

3.1 Data Collection.

The "Heart Disease" dataset has 1025 occurrences and 13 features [16]. Each of these features helps to categorize whether the person is expected to have a heart attack.

Features:

- Age
- Sex (0= Female; 1=Male)
- Chest pain type (Values 0, 1, 2, 3)
- Resting blood pressure
- Serum cholesterol in mg/dl
- Fasting blood sugar > 120 mg/dl (0= False; 1=True)
- Resting electrocardiographic results (values 0,1,2)
- Maximum heart rate achieved
- Exercise induced angina (0=No; 1=Yes)
- Oldpeak = ST depression induced by exercise relative to rest
- The slope of the peak exercise ST segment
- Number of major vessels (0-3) colored by flourosopy
- Thalassemia disease (0 = normal; 1 = fixed defect; 2 = reversable defect)

3.2 Date Preprocessing

Data preprocessing is a vital role in data analytics [17][18]. Because the number of instances is enormous, we begin by selecting 450 at random. We started by looking for missing values but didn't find any. We then searched for outliers. Outliers cannot be recognized since the output is nominal. As a result, all numerical properties must be checked for outliers.

Age, resting blood pressure, cholesterol, maximum heart rate achieved, and oldpeak are the numerical properties. We identified a few outliers, as shown in Table 1.

Table 1. List of outliers

Features	Outliers
Age	Nothing
Resting blood pressure	174, 174, 178, 178, 178, 180, 180, 180, 180, 180, 192, 200, 200, 200
Cholesterol	394, 394, 407, 409, 409, 417, 417, 564, 564
Maximum heart rate achieved	71, 88, 88, 90, 90
Oldpeak	5.6, 5.6, 5.6, 6.2, 6.2

Only the severe outliers were excluded due to the fact that the minor outliers influence the ultimate diagnosis. The interquartile range, or IQR, which is a measure of data dispersion, and Q1, Q3, which represent the lower and upper quartiles, correspondingly, are used to identify the severe outliers in Equations (1) and (2).

$$Q3 + 1.5 \times IQR \tag{1}$$

$$Q1 - 1.5 \times IQR \tag{2}$$

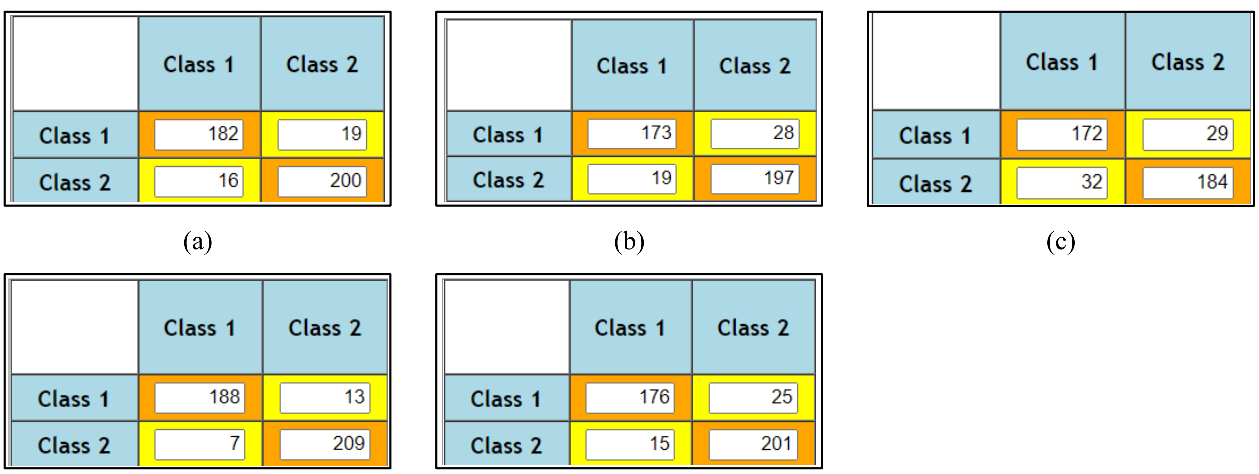
Data points that exceeded the first term were excluded. Likewise, data points below the second term were excluded. Thus, 33 out of the 450 instances were excluded.

4. MODEL DESIGN & RESULTS

Here, five various models will be constructed. Different assessment measures, including accuracy, precision, recall, ROC area, and F-score, will be utilized to decide which is the best.

4.1 Models Before Feature Selection

We used the 10-fold cross validation approach to start. The dataset was divided into ten disjoint subgroups of equal size. Each subset served as the test set, and the remaining 9 subsets were combined to serve as the training set as we developed a classifier. Ten executions of the technique result in ten accuracy values. The average of the 10 accuracies is the final assessed accuracy of learning. Additionally, Weka was used to create an MLP model with 10 hidden layers, 10 neurons, learning rate of 0.1 and the Sigmoid function. The heart disease prediction model was constructed using the five chosen machine learning approaches —MLP, RBF, SVM, KNN, RF— and the results were acquired to arrive at the best final model. The early findings for each model are reported in Table 2 according to the confusion matrices displayed in Fig. 7.



	Class 1	Class 2
Class 1	182	19
Class 2	16	200

(a)

	Class 1	Class 2
Class 1	173	28
Class 2	19	197

(b)

	Class 1	Class 2
Class 1	172	29
Class 2	32	184

(c)

	Class 1	Class 2
Class 1	188	13
Class 2	7	209

(d)

	Class 1	Class 2
Class 1	176	25
Class 2	15	201

(e)

Figure 7. Confusion matrix of (a) MLP, (b) RBF, (c) SVM, (d) KNN, and (e) RF

Table 2. Results and metrics of each model

Model / Metric	MLP	RBF	SVM	KNN	RF
Accuracy	91.6%	88.7%	85.4%	**95.2%**	90.4%
Precision	91.9%	90.1%	84.3%	**96.4%**	92.1%
Recall	90.5%	86.1%	85.6%	**93.5%**	87.6%
F-score	91.2%	88.0%	84.9%	**94.9%**	89.8%
ROC Area	95.2%	92.3%	85.4%	96.1%	**97.6%**

Using Weka, the data above demonstrates that KNN is the best-performing model since it has the greatest F-score and a high ROC area.

4.2 Feature Selection

In some cases, fewer input parameters could improve the model's performance while also reducing the expense of running the model. KNN is the best model for representing the data, thus by removing characteristics that don't contribute to the output, its accuracy may be increased even more. To assess each characteristic's impact, a correlation matrix demonstrating the association between each attribute and the output should be constructed.

The link between the different characteristics and the output was then investigated using the correlation matrix, which was then calculated. The correlation matrix is shown in Figure 9 in which the coefficient denotes the degree and direction of the connection among the variables.

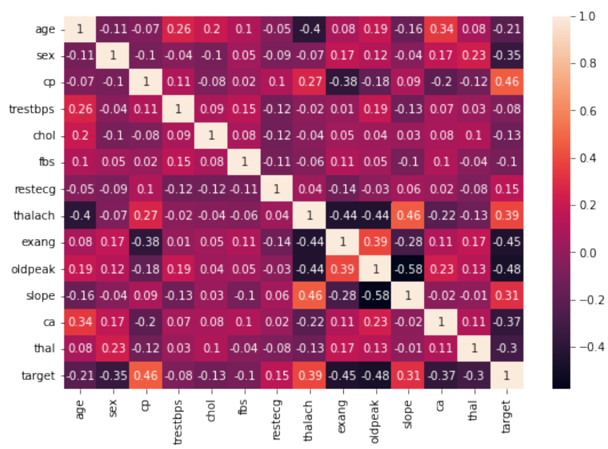


Figure 8. Correlation matrix

According to Figure 9, the characteristics cp, thalach, exang, oldpeak, slope, Ca and sex have the strongest association with the output. These attributes will be utilized to create a new KNN model and evaluate the results. In table 3, the outcomes and metrics are presented.

Table 3. KNN model performance after feature selection

Model / Metric	Accuracy	Precision	Recall	F-score	ROC Area
KNN	92.57%	93.8%	90.5%	92.2%	95.6%

Although there is a little decrease in the performance, this is acceptable given that the model's complexity and computation requirements have been significantly reduced.

5. CONCLUSION

According to particular health parameters, this research seeks to predict the presence of heart disease in individuals using several machine learning techniques.

The study included 5 categorization strategies for building the model. The dataset was extensively checked for severe outliers and missing values after collection. It was also preprocessed to meet the model's needs, going through many stages to find the correlation matrix while utilizing feature selection approaches and the 10-fold cross validation method. Each machine learning method used in the model was trained and evaluated. KNN algorithm had the highest performance with 92.57% accuracy, 92.2% F-measure and ROC Area of 95.6%. The sophisticated relationships between the illness and its symptoms may be extracted thanks to the algorithms applied. Other disorders can also benefit from the use of machine learning algorithms, particularly when more precise medical datasets are produced in the future. By applying more detailed data analysis and trying with different methods, this work's accuracy can be improved.

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